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Problem 1. *Consider the following nonparametric regression model:*

$$\begin{aligned} y_1 &= g(x_1) + \eta + \epsilon_1 \\ y_2 &= h(x_2) + \eta + \epsilon_2 \\ (x_1, x_2) &\perp\!\!\!\perp (\eta, \epsilon_1, \epsilon_2) \end{aligned}$$

Assume further that η, ϵ_1 and ϵ_2 are mutually independent with mean zero, and that the distributions of ϵ_1 and ϵ_2 are identical and symmetric. Discuss the identifiability of the two regression functions and the two distribution functions.

Solution. The regression functions g and h are identifiable from the conditional expectations

$$E[y_1|x_1] = g(x_1), \quad E[y_2|x_2] = h(x_2).$$

Since η, ϵ_1, and ϵ_2 have mean zero and are independent of x_1 and x_2, the conditional expectations identify the regression functions directly. The unobserved heterogeneity η and errors ϵ_t average out conditional on the regressors.

To identify the distributions of η and ϵ_1 (and hence ϵ_2), define the residuals

$$d_1 = y_1 - g(x_1) = \eta + \epsilon_1, \quad d_2 = y_2 - h(x_2) = \eta + \epsilon_2.$$

The joint characteristic function of (d_1, d_2) factors due to mutual independence:

$$\phi_{d_1,d_2}(t_1, t_2) = \phi_\eta(t_1 + t_2)\phi_\epsilon(t_1)\phi_\epsilon(t_2),$$

where ϕ_η and ϕ_ϵ denote the characteristic functions of η and ϵ_1, respectively.

Since ϵ_1 and ϵ_2 are symmetric, ϕ_ϵ is real and even. This property enables recovery of

$$\phi_\epsilon(t) = \sqrt{\phi_{d_1,d_2}(t, -t)},$$

taking the continuous branch with $\phi_\epsilon(0) = 1$. The symmetry ensures that $\phi_\epsilon(-t) = \overline{\phi_\epsilon(t)} = \phi_\epsilon(t)$ for real characteristic functions. Subsequently,

$$\phi_\eta(t) = \frac{\phi_{d_1,d_2}(t, 0)}{\phi_\epsilon(t)}.$$

Both distributions are therefore identified from the observable data through this characteristic function decomposition, provided the characteristic functions are continuous, $\phi_\epsilon(t) \neq 0$ for all t (ensuring the square root is well-defined and the division is valid), and the distributions have finite moments or characteristic functions that do not vanish too rapidly. □

Problem 2. *Suppose the following nonparametric regression model holds:*

$$Y = m(X^*) + U$$

where X^* is endogenous and $m : \mathbb{R} \to \mathbb{R}$ is an unknown function. Consider a reduced form

$$X^* = \pi(Z) + \epsilon.$$

Rather than observing X^*, only an error-ridden version is available:

$$X = X^* + \eta$$

where η represents measurement error. Assume that the density of η, denoted by f_η, is known. Assume that

$$\begin{bmatrix} X^* \\ Z \\ U \end{bmatrix} \perp\!\!\!\perp \eta, \quad E\left[\begin{bmatrix} \epsilon \\ \eta \end{bmatrix} \middle| Z\right] = 0, \quad E[U|\epsilon, Z] = h(\epsilon),$$

and the rank condition $\frac{\partial}{\partial z}\pi(z) \neq 0$ for all $z \in \mathbb{R}$ holds. Derive an expression for the structural derivative $\frac{\partial}{\partial x^*}m(x^*)$ for $x^* \in \mathbb{R}$ in terms of observables.

Solution. We identify the structural derivative through a sequence of steps combining control function methods with deconvolution techniques.

First, we identify the reduced form $\pi(z)$. Since $E[\eta|Z] = 0$ and $E[\epsilon|Z] = 0$, we have

$$E[X|Z = z] = E[X^* + \eta|Z = z] = E[\pi(z) + \epsilon + \eta|Z = z] = \pi(z).$$

Thus $\pi(z)$ is nonparametrically identified from the conditional expectation of X given Z.

Next, we establish that $\epsilon \perp Z$. From the reduced form $X^* = \pi(Z) + \epsilon$ and the assumption $E[\epsilon|Z] = 0$, if we further assume that Z is a valid instrument (which is implicit in the IV setup), then ϵ represents the part of X^* orthogonal to Z. The condition $E[\epsilon|Z] = 0$ combined with the structural assumptions implies $\epsilon \perp Z$. This independence is standard in nonparametric IV models where Z serves as an exogenous instrument.

We now apply the control function approach. Using the control function representation $E[U|\epsilon, Z] = h(\epsilon)$ and the law of iterated expectations,

$$\begin{aligned} E[Y|Z = z] &= E[E[Y|X^*, Z]|Z = z] \\ &= E[m(X^*) + E[U|\epsilon, Z]|Z = z] \\ &= E[m(\pi(z) + \epsilon) + h(\epsilon)|Z = z]. \end{aligned}$$

Since $\epsilon \perp Z$ with density f_ϵ, we obtain

$$E[Y|Z = z] = \int [m(\pi(z) + u) + h(u)] f_\epsilon(u)\, du.$$

Differentiating to isolate the structural derivative, we take the derivative of both sides with respect to z:

$$\begin{aligned} \frac{d}{dz}E[Y|Z = z] &= \int \frac{\partial m(\pi(z) + u)}{\partial x^*} \cdot \pi'(z) \cdot f_\epsilon(u)\, du \\ &= \pi'(z) \int m'(\pi(z) + u) f_\epsilon(u)\, du, \end{aligned}$$

where $\pi'(z) = \frac{d\pi(z)}{dz} \neq 0$ by the rank condition. Similarly, $\frac{d}{dz}E[X|Z = z] = \pi'(z)$. We define the observable ratio

$$\Lambda(z) := \frac{dE[Y|Z = z]/dz}{dE[X|Z = z]/dz} = \int m'(\pi(z) + u) f_\epsilon(u)\, du.$$

Through a change of variables, we let $v = \pi(z)$. Then $\Lambda(\pi^{-1}(v)) =: A(v)$ satisfies

$$A(v) = \int m'(v+u) f_\epsilon(u)\, du = (m' * f_\epsilon)(v),$$

where $*$ denotes convolution. The function $A(v)$ is identified from observables since $E[Y|Z]$ and $E[X|Z]$ are identified from the data, their derivatives are computable under smoothness, the ratio eliminates the unknown $\pi'(z)$, and the change of variables is valid since π is monotone by the rank condition.

To identify the density f_ϵ via deconvolution, observe that for the mismeasured regressor,

$$X - \pi(Z) = X^* + \eta - \pi(Z) = \pi(Z) + \epsilon + \eta - \pi(Z) = \epsilon + \eta.$$

Since $(X^*, Z, U) \perp \eta$ and $\epsilon \perp \eta$, the characteristic function satisfies

$$\phi_{X-\pi(Z)}(t) = E[e^{it(X-\pi(Z))}] = E[e^{it(\epsilon+\eta)}] = \phi_\epsilon(t)\phi_\eta(t).$$

The left-hand side is identified from observables: we estimate $\pi(z)$ from the data, compute residuals $X_i - \hat{\pi}(Z_i)$, and calculate their empirical characteristic function. Since f_η (and hence ϕ_η) is known and $\phi_\eta(t) \neq 0$ for all t (a standard assumption for deconvolution), we recover

$$\phi_\epsilon(t) = \frac{\phi_{X-\pi(Z)}(t)}{\phi_\eta(t)}.$$

Taking the inverse Fourier transform yields f_ϵ.

Finally, we recover the structural derivative via deconvolution. With both $A(v)$ and f_ϵ identified, we solve the convolution equation $A(v) = (m' * f_\epsilon)(v)$ for m'. In Fourier space,

$$\mathcal{F}[A](\omega) = \mathcal{F}[m'](\omega) \cdot \phi_\epsilon(\omega),$$

so

$$\mathcal{F}[m'](\omega) = \frac{\mathcal{F}[A](\omega)}{\phi_\epsilon(\omega)}.$$

Taking the inverse Fourier transform,

$$\frac{\partial m(x^*)}{\partial x^*} = \mathcal{F}^{-1}\left[\frac{\mathcal{F}[A](\omega)}{\phi_\epsilon(\omega)}\right](x^*).$$

This expression is in terms of observables since $A(v)$ is computed from the data and $\phi_\epsilon(\omega) = \phi_{X-\pi(Z)}(\omega)/\phi_\eta(\omega)$ is identified from the data and the known ϕ_η. The procedure requires that $\phi_\epsilon(\omega) \neq 0$ (ensuring well-posedness of deconvolution), sufficient smoothness for differentiation, and appropriate decay conditions for the Fourier transforms to exist. □

Problem 3. *Consider the following linear bivariate regression model:*

$$y_i = \alpha + \beta x_i + u_i, \quad \alpha \in \mathbb{R},\, \beta \in \mathbb{R}$$

with

$$E[u_i|x_i] = 0.$$

Suppose y_i is measured correctly, but x_i is not directly measured. Instead, two mismeasured observations x_i^a and x_i^b are available to the econometrician. That is, IID samples of (y_i, x_i^a, x_i^b), $i = 1, \ldots, n$ are observed.

Define

$$\hat{\beta} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i^a - \bar{x}^a)}{\sum_{i=1}^n (x_i^a - \bar{x}^a)^2}$$

and

$$\tilde{\beta} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i^b - \bar{x}^b)}{\sum_{i=1}^n (x_i^a - \bar{x}^a)(x_i^b - \bar{x}^b)}.$$

(i) Suppose

$$x_i^a = x_i + \eta_i^a, \quad x_i^b = x_i + \eta_i^b,$$

with

$$Cov(\eta_i^a, \eta_i^b) = 0, \quad Cov(u_i, \eta_i^a) = 0, \quad Cov(u_i, \eta_i^b) = 0,$$
$$Cov(x_i, \eta_i^a) = 0, \quad \text{and} \quad Cov(x_i, \eta_i^b) = 0.$$

Are the estimators $\hat{\beta}$ and $\tilde{\beta}$ consistent for β? If not, what are the probability limits?

(ii) Consider now a different model where x_i is binary and Bernoulli:

$$x_i = \begin{cases} 1 & \text{with probability } \lambda \\ 0 & \text{with probability } 1 - \lambda. \end{cases}$$

The assumption $E[u_i|x_i] = 0$ is maintained. The mismeasured binary variables x^a and x^b obey:

$$\begin{aligned} \Pr(x_i^a = 1|x_i = 1) &= \Pr(x_i^b = 1|x_i = 1) = p, \\ \Pr(x_i^a = 0|x_i = 1) &= \Pr(x_i^b = 0|x_i = 1) = 1 - p, \\ \Pr(x_i^a = 1|x_i = 0) &= \Pr(x_i^b = 1|x_i = 0) = q, \\ \Pr(x_i^a = 0|x_i = 0) &= \Pr(x_i^b = 0|x_i = 0) = 1 - q, \end{aligned}$$

and u_i, x_i^a and x_i^b are independent conditional on x_i. Is the estimator $\tilde{\beta}$ consistent for β? If not, what is the probability limit?

Solution.

(i) For $\hat{\beta}$, the probability limit is

$$\operatorname{plim} \hat{\beta} = \frac{\operatorname{Cov}(y, x^a)}{\operatorname{Var}(x^a)} = \frac{\operatorname{Cov}(\beta x + u, x + \eta^a)}{\operatorname{Var}(x + \eta^a)} = \frac{\beta \operatorname{Var}(x)}{\operatorname{Var}(x) + \operatorname{Var}(\eta^a)}.$$

This is the classic attenuation bias result: $\hat{\beta}$ is inconsistent and biased toward zero by the attenuation factor $\lambda_a = \frac{\operatorname{Var}(x)}{\operatorname{Var}(x)+\operatorname{Var}(\eta^a)} < 1$.

For $\tilde{\beta}$, analyze the numerator and denominator:

$$\operatorname{plim} \tilde{\beta} = \frac{\operatorname{Cov}(y, x^b)}{\operatorname{Cov}(x^a, x^b)} = \frac{\operatorname{Cov}(\beta x + u, x + \eta^b)}{\operatorname{Cov}(x + \eta^a, x + \eta^b)} = \frac{\beta \operatorname{Var}(x)}{\operatorname{Var}(x)} = \beta.$$

Since $\operatorname{Cov}(x^a, x^b) = \operatorname{Var}(x)$ (the measurement errors are uncorrelated with each other and with the true x), $\tilde{\beta}$ is consistent for β.

(ii) In the binary case, $E[x] = \lambda$ and $\operatorname{Var}(x) = \lambda(1-\lambda)$. For the misclassified variables,

$$E[x^a | x = 1] = p, \quad E[x^a | x = 0] = q, \quad E[x^a] = \lambda p + (1-\lambda) q.$$

The covariance between y and x^b is

$$\operatorname{Cov}(y, x^b) = E[y x^b] - E[y]E[x^b] = \beta E[x x^b] - \beta \lambda E[x^b].$$

Computing $E[x x^b] = E[x \cdot E[x^b | x]] = E[x \cdot (p \cdot \mathbf{1}_{\{x=1\}} + q \cdot \mathbf{1}_{\{x=0\}})] = \lambda p$ yields

$$\operatorname{Cov}(y, x^b) = \beta(\lambda p - \lambda(\lambda p + (1-\lambda) q)) = \beta \lambda (1-\lambda)(p-q).$$

The covariance between x^a and x^b is

$$\operatorname{Cov}(x^a, x^b) = E[x^a x^b] - E[x^a]E[x^b].$$

Computing $E[x^a x^b] = E[E[x^a x^b | x]] = E[E[x^a | x] E[x^b | x]] = E[(p \cdot \mathbf{1}_{\{x=1\}} + q \cdot \mathbf{1}_{\{x=0\}})^2] = \lambda p^2 + (1-\lambda) q^2$ yields

$$\operatorname{Cov}(x^a, x^b) = \lambda p^2 + (1-\lambda) q^2 - (\lambda p + (1-\lambda) q)^2 = \lambda(1-\lambda)(p-q)^2.$$

Therefore, the probability limit is

$$\operatorname{plim} \tilde{\beta} = \frac{\operatorname{Cov}(y, x^b)}{\operatorname{Cov}(x^a, x^b)} = \frac{\beta \lambda (1-\lambda)(p-q)}{\lambda(1-\lambda)(p-q)^2} = \frac{\beta}{p-q}.$$

Hence $\tilde{\beta}$ is inconsistent unless $p - q = 1$, which would correspond to no misclassification. The inconsistency arises because misclassification creates both attenuation and contamination bias in the binary case.

□

Problem 4. *Let $n \to \infty$ in verifying convergence properties.*

(i) Consider the following $\mathbb{R}$-valued random variable:

$$X_n = \begin{cases} \mathrm{N}(\theta, 1) & \text{with probability } \frac{n-1}{n} \\ \theta + n & \text{with probability } \frac{1}{n} \end{cases}$$

where θ is a constant.

(a) *Does X_n converge in distribution? If so, what is the limiting distribution? First use Definition 1 of convergence in distribution, then answer the same question again using Definition 2.*

(b) *Does the mean of X_n converge to that of X? Discuss the result in relation to the statement of the Helly-Bray Theorem.*

(ii) *Consider the following $\mathbb{R}$-valued random variable:*

$$X_n = \begin{cases} \mathrm{N}(0,1) & \text{with probability } \frac{1}{2} \\ \mathrm{N}(0,\frac{1}{n}) & \text{with probability } \frac{1}{2} \end{cases}$$

(a) *Use both Definition 1 and Definition 2 to verify that X_n converges in distribution, i.e., $X_n \xrightarrow{d} X$ for some r.v. X. What is the limiting distribution?*

(b) *Let F_n and F denote the CDFs of X_n and X, respectively. Let $g(x) = \mathbf{1}\{x < 0\}$. Does*

$$\int g\, dF_n \to \int g\, dF$$

hold? What if $g(x) = \mathbf{1}\{x < \epsilon\}$, $\epsilon > 0$? Discuss the answer in view of the statement of the stronger version of the Helly-Bray theorem.

Solution.

(a) Using Definition 1 (convergence of CDFs), for any $x \in \mathbb{R}$,

$$F_n(x) = \frac{n-1}{n}\Phi(x-\theta) + \frac{1}{n} \cdot \mathbf{1}\{x \geq \theta + n\} \to \Phi(x-\theta)$$

as $n \to \infty$, so $X_n \xrightarrow{d} N(\theta, 1)$. The point mass at $\theta + n$ vanishes asymptotically because its probability $1/n \to 0$.

Using Definition 2 (convergence of expectations for bounded continuous functions), for any bounded continuous function g,

$$E[g(X_n)] = \frac{n-1}{n}E[g(N(\theta,1))] + \frac{1}{n}g(\theta+n) \to E[g(N(\theta,1))],$$

since $g(\theta + n)$ is bounded and multiplied by $1/n \to 0$. Thus, $X_n \xrightarrow{d} N(\theta, 1)$.

(b) The mean of X_n is

$$E[X_n] = \frac{n-1}{n}\theta + \frac{1}{n}(\theta + n) = \theta + 1.$$

This converges to $\theta + 1$, while the mean of the limiting distribution $N(\theta, 1)$ is θ. This does not violate the Helly-Bray theorem, which only guarantees convergence of expectations for bounded continuous functions. Since $g(x) = x$ is unbounded, the theorem does not apply.

(ii) (a) Using Definition 1, for any $x \in \mathbb{R}$,

$$F_n(x) = \frac{1}{2}\Phi(x) + \frac{1}{2}\Phi(x\sqrt{n}).$$

For $x < 0$, we have $\Phi(x\sqrt{n}) \to \Phi(-\infty) = 0$, so $F_n(x) \to \frac{1}{2}\Phi(x)$. For $x > 0$, we have $\Phi(x\sqrt{n}) \to \Phi(\infty) = 1$, so $F_n(x) \to \frac{1}{2}\Phi(x) + \frac{1}{2}$. At $x = 0$, there is a discontinuity: $F_n(0) = \frac{1}{2}\Phi(0) + \frac{1}{2}\Phi(0) = \frac{1}{2}$, while

$$F(0) = \lim_{x \to 0^+} F(x) = \frac{1}{2}\Phi(0) + \frac{1}{2} = \frac{3}{4}.$$

Since convergence in distribution requires convergence only at continuity points of the limit CDF, and 0 is a discontinuity point of F, the result $X_n \xrightarrow{d} X$ still holds, where X has the limiting distribution

$$F(x) = \begin{cases} \frac{1}{2}\Phi(x) & \text{if } x < 0, \\ \frac{1}{2}\Phi(x) + \frac{1}{2} & \text{if } x \geq 0. \end{cases}$$

This is a mixture of $N(0,1)$ with probability $1/2$ and a point mass at 0 with probability $1/2$.

Using Definition 2, for any bounded continuous function g,

$$E[g(X_n)] = \frac{1}{2}E[g(N(0,1))] + \frac{1}{2}E[g(N(0,1/n))].$$

Since $N(0,1/n) \xrightarrow{p} 0$ and g is continuous, we have $g(N(0,1/n)) \xrightarrow{p} g(0)$. By boundedness and convergence in probability, $E[g(N(0,1/n))] \to g(0)$. Thus,

$$E[g(X_n)] \to \frac{1}{2}E[g(N(0,1))] + \frac{1}{2}g(0) = E[g(X)],$$

confirming $X_n \xrightarrow{d} X$.

(b) For $g(x) = \mathbf{1}\{x < 0\}$,

$$\begin{aligned} \int g \, dF_n &= P(X_n < 0) \\ &= \frac{1}{2}P(N(0,1) < 0) + \frac{1}{2}P(N(0,1/n) < 0) \\ &= \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}. \end{aligned}$$

For the limiting distribution (50% $N(0,1)$ and 50% point mass at 0),

$$\int g \, dF = P(X < 0) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 0 = \frac{1}{4}.$$

Since $\frac{1}{2} \neq \frac{1}{4}$, convergence does NOT hold for this discontinuous function.

For $g(x) = \mathbf{1}\{x < \epsilon\}$ with $\epsilon > 0$,

$$\begin{aligned}\int g\, dF_n &= F_n(\epsilon^-) \\ &= \frac{1}{2}\Phi(\epsilon) + \frac{1}{2}\Phi(\epsilon\sqrt{n}) \\ &\to \frac{1}{2}\Phi(\epsilon) + \frac{1}{2}\cdot 1 \\ &= F(\epsilon^-).\end{aligned}$$

Since $\epsilon > 0$ is a continuity point of F, convergence holds. This illustrates the Portmanteau Theorem (stronger version of Helly-Bray): convergence in distribution is equivalent to $\int g\, dF_n \to \int g\, dF$ for all bounded functions g that are continuous F-almost everywhere. The function $\mathbf{1}\{x < 0\}$ is discontinuous at 0, which is an atom of the limit distribution (has positive mass), so convergence fails as expected.

□

Problem 5. *Define*

$$X_n = \begin{cases} \mathrm{N}(\theta, \frac{1}{n}) & \textit{with probability } \frac{n-1}{n} \\ \theta + \sqrt{n} & \textit{with probability } \frac{1}{n} \end{cases}$$

where θ is a constant. Does X_n converge to θ in:

(a) probability;

(b) L_1;

(c) L_2?

Solution.

(a) For convergence in probability, we must show that for any $\epsilon > 0$, $\Pr(|X_n - \theta| > \epsilon) \to 0$. Observe that

$$\Pr(|X_n - \theta| > \epsilon) = \frac{n-1}{n}\Pr(|N(0, 1/n)| > \epsilon) + \frac{1}{n}\Pr(|\sqrt{n}| > \epsilon).$$

For the first term,

$$\Pr(|N(0, 1/n)| > \epsilon) = \Pr(|N(0,1)| > \epsilon\sqrt{n}) \to 0$$

as $n \to \infty$, since the tails of the standard normal distribution decay exponentially. Multiplying by $(n-1)/n \to 1$ preserves this convergence. For the second term, $\frac{1}{n} \cdot \mathbf{1}\{\sqrt{n} > \epsilon\} \to 0$ since $\frac{1}{n} \to 0$. Thus, $X_n \xrightarrow{p} \theta$.

(b) For convergence in L_1, we compute

$$\begin{aligned}E[|X_n - \theta|] &= \frac{n-1}{n}E[|N(0, 1/n)|] + \frac{1}{n}E[|\sqrt{n}|] \\ &= \frac{n-1}{n}\sqrt{\frac{1}{n}}E[|N(0,1)|] + \frac{1}{n}\sqrt{n} \\ &= \frac{n-1}{n}\cdot\frac{1}{\sqrt{n}}\cdot\sqrt{\frac{2}{\pi}} + \frac{1}{\sqrt{n}}.\end{aligned}$$

As $n \to \infty$, the first term behaves like $\frac{1}{\sqrt{n}} \to 0$ and the second term $\frac{1}{\sqrt{n}} \to 0$. Thus, $E[|X_n - \theta|] \to 0$, so $X_n \xrightarrow{L_1} \theta$.

(c) For convergence in L_2, we compute

$$\begin{aligned} E[(X_n - \theta)^2] &= \frac{n-1}{n} E[(N(0, 1/n))^2] + \frac{1}{n} E[(\sqrt{n})^2] \\ &= \frac{n-1}{n} \cdot \frac{1}{n} + \frac{1}{n} \cdot n \\ &= \frac{n-1}{n^2} + 1. \end{aligned}$$

As $n \to \infty$, this converges to $0 + 1 = 1 \neq 0$. Thus, X_n does not converge to θ in L_2. Although the outlier at $\theta + \sqrt{n}$ occurs with vanishing probability, its squared deviation grows linearly with n, creating a non-vanishing second moment.

□

Problem 6. *(i) Show that if $Y_n \xrightarrow{p} Y$ then $Y_n \xrightarrow{d} Y$ (here Y is random).*

(ii) Consider a random variable X and a random sequence $\{W_n\}_{n=1,2,\ldots}$ (all scalar valued). Suppose $X \sim N(0,1)$ and $W_n \xrightarrow{p} 0$. Define:

$$Y_n = X + W_n, \quad Z_n = X, \quad n = 1, 2, \ldots$$

Trivially, $Y_n \xrightarrow{p} X$ and $Z_n \xrightarrow{d} X$. What is the joint limit distribution of $\binom{Y_n}{Z_n}$?

(iii) Consider the same setting as (ii), but define instead

$$Y_n = X + W_n, \quad Z_n = -X, \quad n = 1, 2, \ldots$$

Again, $Y_n \xrightarrow{p} X$ and $Z_n \xrightarrow{d} X$ (the latter holds by symmetry of $N(0,1)$). What is the joint limit distribution of $\binom{Y_n}{Z_n}$? In view of this example, does the following result hold? "Suppose $Y_n \xrightarrow{p} Y$ and $Z_n \xrightarrow{d} Z$, then $\binom{Y_n}{Z_n} \xrightarrow{d} \binom{Y}{Z}$."

Solution.

(i) To show $Y_n \xrightarrow{d} Y$, we use Definition 2: for every bounded continuous function g, we must show $E[g(Y_n)] \to E[g(Y)]$.

Since $Y_n \xrightarrow{p} Y$ and g is continuous, by the Continuous Mapping Theorem, $g(Y_n) \xrightarrow{p} g(Y)$. We now show that convergence in probability of bounded random variables implies convergence of expectations. Let $|g(x)| \leq M$ for all x. For any $\epsilon > 0$, choose $\delta > 0$ (to be determined). Then

$$\begin{aligned} |E[g(Y_n)] - E[g(Y)]| &\leq E[|g(Y_n) - g(Y)|] \\ &= E[|g(Y_n) - g(Y)| \cdot \mathbf{1}\{|Y_n - Y| > \delta\}] + E[|g(Y_n) - g(Y)| \cdot \mathbf{1}\{|Y_n - Y| \leq \delta\}]. \end{aligned}$$

For the first term,

$$E[|g(Y_n) - g(Y)| \cdot \mathbf{1}\{|Y_n - Y| > \delta\}] \leq 2M \cdot P(|Y_n - Y| > \delta).$$

Since $Y_n \xrightarrow{p} Y$, we have $P(|Y_n - Y| > \delta) \to 0$ as $n \to \infty$. Thus for n large enough, this term is less than $\epsilon/2$.

For the second term, since g is continuous on $\mathbb{R}$ and bounded, it is uniformly continuous on any bounded interval. The random variables Y and Y_n have tight distributions (they converge in probability, hence are tight), so with probability approaching 1, both lie in some bounded interval $[-K, K]$. On this interval, by uniform continuity, there exists $\delta > 0$ such that $|y_n - y| \leq \delta$ implies $|g(y_n) - g(y)| < \epsilon/2$. Therefore,

$$E[|g(Y_n) - g(Y)| \cdot \mathbf{1}\{|Y_n - Y| \leq \delta\}] < \epsilon/2.$$

Combining both terms, $|E[g(Y_n)] - E[g(Y)]| < \epsilon$ for n sufficiently large. Since ϵ was arbitrary, $E[g(Y_n)] \to E[g(Y)]$, establishing $Y_n \xrightarrow{d} Y$.

Alternatively, this result follows directly from the standard theorem that convergence in probability implies convergence in distribution, which in turn is equivalent to the convergence $E[g(Y_n)] \to E[g(Y)]$ for all bounded continuous g by the bounded convergence theorem for probability measures.

(ii) Here $Y_n = X + W_n \xrightarrow{p} X$ and $Z_n = X$. For the joint distribution, consider any bounded continuous function $g : \mathbb{R}^2 \to \mathbb{R}$. Then

$$E[g(Y_n, Z_n)] = E[g(X + W_n, X)].$$

Since $W_n \xrightarrow{p} 0$ and g is continuous, $g(X + W_n, X) \xrightarrow{p} g(X, X)$. Since g is bounded, by the argument in part (i), $E[g(Y_n, Z_n)] \to E[g(X, X)]$. Thus, the joint limit distribution is that of $\binom{X}{X}$, where $X \sim N(0, 1)$.

(iii) Now $Y_n = X + W_n \xrightarrow{p} X$ and $Z_n = -X \xrightarrow{d} X$ (by symmetry of $N(0, 1)$). For the joint distribution, consider the same bounded continuous function g:

$$E[g(Y_n, Z_n)] = E[g(X + W_n, -X)] \to E[g(X, -X)],$$

by the same reasoning. Thus, the joint limit distribution is $\binom{X}{-X}$, where $X \sim N(0, 1)$.

This counterexample shows that the statement "Suppose $Y_n \xrightarrow{p} Y$ and $Z_n \xrightarrow{d} Z$, then $\binom{Y_n}{Z_n} \xrightarrow{d} \binom{Y}{Z}'$ is **false** in general. Here, $Y_n \xrightarrow{p} X$ and $Z_n \xrightarrow{d} X$, but the joint limit is $\binom{X}{-X}$, not $\binom{X}{X}$. The issue is that convergence in distribution of the marginal distributions does not guarantee convergence of the joint distribution. For joint convergence to $\binom{Y}{Z}$, we would need joint weak convergence or additional conditions such as asymptotic independence. Convergence in probability of one component and convergence in distribution of another is insufficient without knowing the joint behavior.

□